

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 8: Low-Altitude Radio Altimeter (FMCW)

Syllabus Source: CAE Oxford (pp. 269-298) | Pass Target: >= 90% AviationExam

1. FREQUENCY MODULATED CONTINUOUS WAVE (FMCW) RADAR

The Radio Altimeter (Rad Alt) measures the precise absolute geometric clearance of aircraft main landing gear wheels above underlying terrain from 0 to 2,500 ft AGL:

- Frequency Band: Operates in the SHF aeronautical radionavigation band between 4.2 GHz and 4.4 GHz (typically center frequency 4.3 GHz).
- FMCW Modulation Technique: Instead of high-power pulses, the transmitter outputs a CONTINUOUS signal whose carrier frequency is swept linearly up and down at a constant modulation rate (triangular or sinusoidal frequency sweep).
- Beat Frequency Principle: Signal transmitted downward reflects off the terrain and returns to the receiver antenna after a time delay: $\Delta t = (2 \times \text{Height}) / c$. During this transit time, the transmitter has swept to a new frequency! Mixing the received signal with the current transmitted signal produces a BEAT FREQUENCY (Δf) that is DIRECTLY PROPORTIONAL to terrain wheel height!

2. INTEGRATION, DECISION HEIGHT & AUTOLAND GUIDANCE

Height Regime	Operational Function & Cockpit Indications	Interfaced Aircraft Systems
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Above 2,500 ft AGL	Radio altimeter indication is INHIBITED or flagged OFF. Pointer parked behind mask; digital readout blanked.	No ground interaction; GPWS radar terrain clearance modes inactive.
2,500 ft down to 50 ft	Readout appears on PFD (round dial or digital tape). Dial expansion: 0-500 ft scale is expanded for high visual precision.	Feeds GPWS Modes 1-4; arms Autoland flare and rollout modes.
Decision Height (DH)	Pilot dials DH bug on EFIS control panel (e.g. 200 ft for CAT I, 100 ft or 50 ft for CAT II).	At DH: Yellow 'DH' annunciates on PFD and synthetic voice announces 'DECISION' or 'MINIMUMS'!
Below 50 ft (Autoland)	Crucial input to autopilot autoland computers.	At 50-30 ft: Triggers autopilot FLARE mode (throttles retard to idle, pitch increases smoothly for touchdown). Synthetic voice callouts: '50, 40, 30, 20, 10, RETARD'.

[!] EXAM TRAP (AVIATIONEXAM) - Radio Altimeter Zero Height Calibration

- The Radio Altimeter antennas are mounted on the fuselage underside. - Calibration Datum: In a level flight landing attitude, when the main landing gear tyres touch the runway concrete, the radio altimeter MUST INDICATE EXACTLY ZERO FEET! - Antenna cable delay lines are built into the receiver harness to compensate for the physical distance between antennas and main wheels.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Radio Altimeter vs Pressure Altimeter**

Rule 1: Pressure altimeter measures static pressure -> Altitudes (MSL) and Flight Levels. Rule 2: Radio altimeter transmits FMCW at 4.2-4.4 GHz -> Absolute height above ground (AGL). Rule 3: Radio altimeter operates ONLY between 0 and 2,500 ft AGL. Above 2,500 ft, pointer is parked or flag displays.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What modulation principle is utilized by a modern low-altitude radio altimeter?**

- [A]: Amplitude Modulation (AM)
- [B]: Frequency Modulated Continuous Wave (FMCW) where the beat frequency between transmitted and reflected waves is proportional to wheel height
- [C]: Pulsed primary radar with 10 microsecond pulses
- [D]: Phase shift keying

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Low-altitude radio altimeters use FMCW. Comparing the transmitted swept frequency against the delayed reflected return frequency produces a beat frequency directly proportional to terrain height.

Q2: At what radio altitude does the radio altimeter indication normally become active on a transport aircraft Prior to landing?

- [A]: 10,000 ft AGL
- [B]: 5,000 ft AGL
- [C]: 2,500 ft AGL
- [D]: 500 ft AGL

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Commercial radio altimeters operate from 0 to 2,500 ft AGL. Above 2,500 ft, the indication is blanked or masked on modern electronic displays.

Q3: When an aircraft is in the landing flare with its main gear wheels touching the runway surface, what should the radio altimeter display?

- [A]: Negative 5 ft
- [B]: Exactly ZERO feet
- [C]: Antenna height (approx 10 ft)
- [D]: Aerodrome elevation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Radio altimeters are calibrated with an internal electronic delay line so that with landing gear extended in touchdown attitude, the display reads exactly 0 ft when the tyres touch down.